

Arnav Kamath

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EDUCATION

University of California, Los Angeles | Los Angeles, CA

Sep. 2026 - Jun. 2030

Computer Science, B.S.

- **Relevant Coursework:** Data Structures and Algorithms (C++), Computer Organization and Architecture, Discrete Mathematics, Linear Algebra, Differential Equations

Foothill High School | Pleasanton, CA

Aug. 2022 - May 2026

- **Relevant Coursework:** Multivariable Calculus, AP Calculus BC, AP Computer Science A, AP Statistics, Honors Digital Electronics, Honors Computer Integrated Manufacturing, Honors Principles of Engineering
- **Activities:** CS Club Co-President, FalconHacks Co-President, Science Olympiad Engineering Captain, Varsity Badminton Captain

EXPERIENCE

DeWeese Lab, University of California, Berkeley | Berkeley, CA

Mar. 2025 - Jan. 2026

Research Intern

- Developed regression-based monocular depth estimation model, achieving an average mean squared error of $\sim 0.05 \text{ m}^2$ over a subset of the NYU Depth V2 dataset while remaining fully interpretable vs. CNN baselines
- Paper titled "Interpretable Monocular Depth Estimation Using Patch-Based Polynomial Regression" under review for publication in the International Journal of High School Research

SynergyPlus | Pleasanton, CA

Aug. 2025 - Dec. 2025

Software Engineering Intern

- Helped architect and deploy a full-stack gradebook and grade-calculation platform that syncs with existing grading software using React, HTML/CSS, and StudentVue API
- Expanded user base, serving 4,500+ students and teachers across multiple school districts

PROJECTS

RepoSpace - AI Repository Intelligence System | Google AI Agents Capstone Challenge

July 2026

- Utilized Google Gemini and the Google Agent Development Kit (ADK) to build a modular agentic framework that ingests Github Repositories and can aid with inquiries and feature development
- Capable of repository indexing, semantic code retrieval, project structure analysis, and implementation assistance. Incorporated Retrieval-Augmented Generation (RAG) and Model Context Protocol (MCP).

Autonomous Driving Algorithm | UC Berkeley ROAR Competition

Aug. 2025

- Developed an open-source autonomous control algorithm which uses PID and a waypoint-following pipeline for generating steering correction and throttle/braking control commands from real-time vehicle environment
- Received 4th place award, following rigorous evaluation using the CARLA simulator

TECHNICAL SKILLS

Languages: Python, Java, HTML/CSS, JavaScript, Lua/Luau

Tools, Frameworks, Libraries: Pandas, Numpy, Scikit-learn, Keras, Matplotlib, PyTorch, TensorFlow, OpenCV, React, Jupyter Notebook, Git, GitHub, Google Antigravity, Google Agent Development Kit (ADK), Claude Code, CAD Design

Certifications: Google AI Agents